

Recurrence Intervals for Great Earthquakes Of the Past 3,500 Years at Northeastern Willapa Bay, Washington

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*A geologic history of earthquakes
at the Cascadia subduction zone*

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brackish intertidal diatoms indices transitional between low marsh and high marsh, these indices are due to probably allochthonous valves of heavily silicified and easily reworked tidal flat diatoms (p. 48). Diatoms in soil U accord with vascular-plant evidence that soil U represents a slightly lower-level high marsh than do soils S and Y (p. 48).

Transitions from tidal flat to low marsh can be inferred both from diatom assemblages and from vascular-plant fossils in mud above soils S, U, and Y at the Oyster locality (fig. 20). Each transition is marked in outcrop by the first appearance of *Triglochin maritimum* rhizomes in soft gray mud above the buried soil. The mud lacks other vascular-plant fossils in growth position but contains rhythmic layers of organic mud ascribed to filamentous algae found today mainly at the transition between tidal flat and low marsh (p. 48). Diatom assemblages independently suggest that mud below the rhizomes represents a tidal flat or shallow subtidal conditions. However, the diatom assemblages, as summarized by the brackish intertidal diatom index, place the transition to low marsh several tenths of a meter above the lowest *Triglochin maritimum* rhizomes. This discrepancy may reflect patchy colonization of the tidal flat, which could have supported dominantly tidal-flat assemblages of diatoms until *Triglochin maritimum* patches fully coalesced. The discrepancy may also result from errors inherent in using diatom assemblages to estimate the point of transition from tidal flat to low marsh (Hemphill-Haley, 1995b).

Inferences from rhizomes and diatoms differ somewhat for a tidal-flat-to-low-marsh transition between soils J and L at the Redtail locality (fig. 15). Diatoms found with the lowest *Triglochin maritimum* rhizomes (at depth 310 cm) include such low-marsh species as *Caloneis westii* and *Diploneis interrupta*. However, the assemblage contains many more valves of the high-marsh diatom *Cosmioneis pusilla* var. 1. These valves, which could have been reworked from high marsh that may have fringed the upland beside the Redtail locality (fig. 7), push the diatom index into the high-marsh range (fig. 15).

RADIOCARBON DATING

Radiocarbon ages have been measured on samples of wood, peat, and rhizomes from the Oyster, Redtail, Pool, Jensen, and Sewer localities. Most of the ages were measured by liquid scintillation methods and were calculated for an assumed fractionation ($\delta^{13}\text{C}$) of -25 parts per thousand (‰) (table 7, lab code Beta). Other samples were dated by proportional gas methods (lab codes QL and USGS). $\delta^{13}\text{C}$ was measured for all the QL ages and was assumed to be -25 ‰ for the USGS ages. Still other samples were dated by accelerator mass spectrometry (RIDDL, GX) and calculated with measured $\delta^{13}\text{C}$.

KINDS OF YEARS

Table 7 and the outcrop diagrams give ages in conventional radiocarbon years before A.D. 1950 (^{14}C yr B.P.). In the text and in interpretive diagrams, however, we mainly use calendric years before A.D. 2,000 (years ago). We converted radiocarbon ages to calendric-year ranges by means of the calibration data of Stuiver and Becker (1993) and version 3.0.3 of the radiocarbon calibration program of Stuiver and Reimer (1993).

We do not use "cal yr B.P.," the standard unit for calibrated radiocarbon ages (Stuiver and Pearson, 1986), because the "B.P." connotes "before present" but denotes "before A.D. 1950." Thus we describe A.D. 1700 as about 300 years ago, not as 250 cal yr B.P.

ANALYTICAL ERRORS IN DATING

The error quoted with each age, as listed in table 7, does not necessarily equal one standard deviation of the age measurement. Such underestimation of analytical uncertainties is common in radiocarbon dating (Scott and others, 1990). The quoted error probably exceeds 0.6–0.8 standard deviation for the QL ages (Stuiver and Pearson, 1986) but may approximate 1.0 standard deviation for the GX ages (Nelson and others, 1995). We lack such figures for the Beta, RIDDL, and USGS ages.

Figure 32 shows two calendric-age ranges for each ^{14}C age. The narrower range equals the 95-percent confidence interval if the error quoted by the lab equals one standard deviation of the age measurement. The wider range incorporates a published, maximum error multiplier of 1.6 for ages measured at the University of Washington (lab code QL; Stuiver and Pearson, 1986) and an assumed and probably excessive error multiplier of 2.0 for the other ages. In other illustrations and in the text we use the wider range, and in the text we commonly use a still broader range rounded outward to the century.

GEOLOGIC ERRORS IN DATING

The material dated formed before or after the inferred earthquakes that we sought to date. For several samples of forest-floor litter this difference is probably negligible (fig. 32, "closely limiting" ages for soils J and N), and for some tree-root samples the difference has been corrected by means of tree-ring counts (Atwater and others, 1991) (fig. 32, "closely limiting" ages for soils S and Y). For other samples, particularly of peat, the difference may amount to decades or centuries (Nelson, 1992b). Arrows in figure 32 identify ages subject to such differences. These include ages on small woody roots that may have died long before an inferred earthquake (arrow pointing upward), ages on rhizomes that may have formed long after the earthquake (arrow pointing downward), and ages on peat and sticks that

18) and away from uplands at the Redtail and Heron localities (figs. 13, 25).

The thickness of intertidal mud above a buried soil can serve as a rough indicator of the amount of coseismic subsidence because subsidence makes room for tides to deposit mud on the soil. The thickness of mud is unlikely to equal the amount of coseismic subsidence, however, if tidal inundation subsequently decreases during interseismic uplift or increases during interseismic subsidence (Darioenzo and Peterson, 1990; Hemphill-Haley, 1995b, p. 377; Guilbault and others, 1995).

NUMERICAL ESTIMATES OF SUBSIDENCE

We do not compare structural setting with numerical estimates of subsidence because, at present, such estimates would be unreliable for the South Bend antiform. Estimates of coseismic subsidence at the Cascadia subduction zone are best made at sites studied carefully for microscopic fossils (Mathewes and Clague, 1994; Hemphill-Haley, 1995b; Guilbault and others, 1995; Nelson and others, 1996a, b; Shennan and others, 1996). Microfossils have not yet been studied for most soils on the South Bend antiform. However, they have been studied abundantly for soils off the antiform, along the Niawiakum River.

Subsidence of at least 1 m was previously inferred for soil Y at the Redtail and Oyster localities, mainly on the basis of diatom assemblages (Hemphill-Haley, 1995b). Here we extend this estimate to most soils at the Redtail and Oyster localities. The estimates are based on inferred differences in depositional environment between buried soils and the mud immediately above them (figs. 15, 20). We inferred the environmental differences from stratigraphy and from species of vascular plants and diatoms that are vertically zoned in modern tidal wetlands (figs. 8, 11, 12).

Subsidence of at least 1 m is probably recorded by soils J, L, N, S, U, and Y at the Redtail locality because all these soils probably contain high marsh or upland assemblages of diatoms and because mud above each of them contains diatom assemblages suggestive of widely unvegetated mudflats (p. 29). The modern difference between these environments is rarely less than 0.5 m and commonly exceeds 1 m (fig. 8, parameter D). The evidence for at least 1 m of subsidence is strongest for soils S and Y, whose spruce roots corroborate diatom evidence for conditions transitional to upland (p. 29).

The Oyster locality also has strong evidence that soils S and Y record at least 1 m of subsidence (p. 48). This evidence consists of high-marsh plants rooted in the soils, algal lamination and *Triglochin maritimum* rhizomes in mud above the soils, and diatom assemblages compatible with these features.

Only soil W has evidence against subsidence greater than 1 m. At the Oyster locality this soil represents a tidal marsh above the lower limit of *Triglochin maritimum* but probably below the transition from low marsh to high marsh

(p. 48). Moreover, after soil W subsided it did not remain a tidal flat for long, as judged from the thickness of mud dominated by tidal-flat diatoms (p. 52). In that case it may have subsided only through the vertical range of low marsh, or as little as about 0.5 m.

COEVALITY OF LAND-LEVEL CHANGE

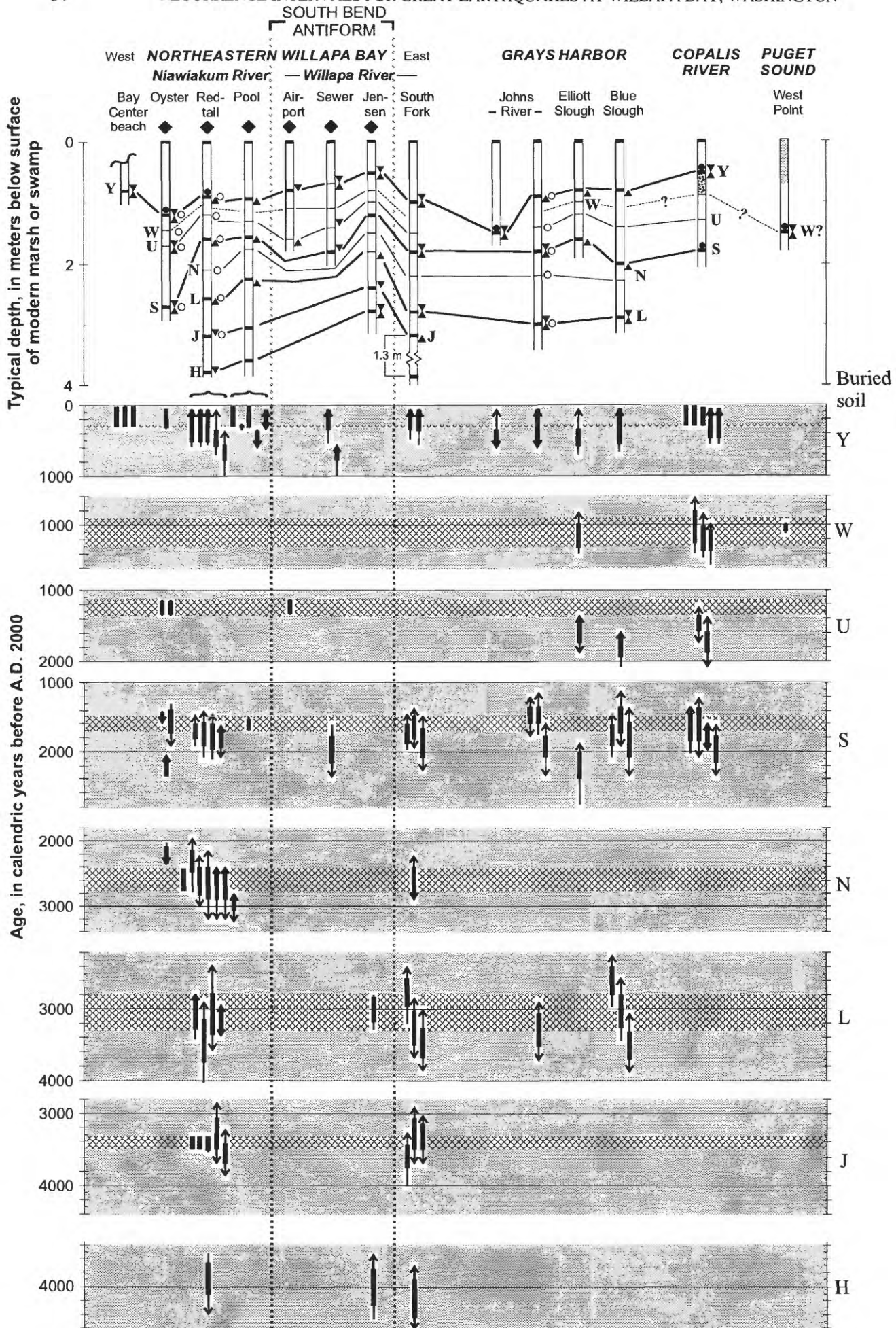
Buried soils of the past 2,000 years on the South Bend antiform do not appear to differ—either in stratigraphic sequence or in radiocarbon age—from buried soils at nearby sites off the antiform (fig. 32). All these soils appear to be part of a regionally correlative sequence previously reported from many estuarine localities along the southern 100 km of the Washington coast (Atwater, 1992). In this sequence, two bold soils (herein, soils S and Y) bracket one or two faint soils (U and W). Soils S and Y commonly contain more fossil tree roots than do soils U and W, and only soil Y is commonly covered by a tsunami deposit at Willapa Bay and Grays Harbor. Ages determined by radiocarbon dating (chiefly for soils S and Y; Atwater, 1992) and by tree-ring pattern matching (soil Y only; Yamaguchi and others, 1997) are consistent with correlations inferred from these various stratigraphic clues. Additional radiocarbon dating of soils in the stratigraphic position of soils S, U, and Y is consistent with correlation between a typical southern-Washington sequence of buried soils at the Naselle River (fig. 5) and a sequence of buried soils at Netarts Bay, Oregon (fig. 1) (Nelson and others, 1996b, their fig. 6).

Buried soils 2,000–3,500 years old also show no major differences in stratigraphy or approximate age between the South Bend antiform and the handful of sites at which such soils have been dated in southern coastal Washington. The four lowest soils on the antiform at the Jensen locality probably correlate with soils H, J, L, and N off the antiform at the Redtail locality (p. 67–69). Soils low in the section at the Jensen locality probably also correlate with soils off the anticline at Willapa Bay (South Fork locality) and Grays Harbor (Johns River and Blue Slough localities) (fig. 32).

POSSIBLY UNUSUAL CHANGES IN LAND LEVEL 1,000–1,100 YEARS AGO

The subsidence marked by soil W at Willapa Bay may have coincided, to the nearest century or less, with unusual faulting in the North America plate. Such faulting probably occurred some 50 km to the north, at the Copalis River. There, vented sand covers a soil, possibly correlative with soil W (fig. 32), that underwent little or no subsidence at the time of venting. This lack of subsidence implies localized deformation in the North America plate because the venting, which is not readily explained by shaking-induced liquefaction, probably resulted from faulting or folding of an aquifer beneath the Copalis River (Atwater, 1992, p. 1917, "Note added in proof"). Such faulting or folding could have been part of widespread movement on structures high in the

RECURRENCE INTERVALS FOR GREAT EARTHQUAKES AT WILLAPA BAY, WASHINGTON



← **Figure 32.** Inferred correlations among buried Holocene soils at estuaries in southern coastal Washington and at Puget Sound. Individual age ranges calculated from radiocarbon ages in Atwater and others (1991), Atwater (1992), Atwater and Moore (1992), and table 7 of this report. Vertically ruled bands denote likely age ranges explained in section "Number and Ages of Earthquakes" (p. 91). Age range for soil Y at Oyster locality corresponds to weighted mean age of 161 ± 15 ^{14}C yr B.P.

EXPLANATION

Columnar sections

- ◆ **Section described in this report**—Column generalized from outcrop drawing (figs. 13, 18, 24, 27, 29, and 31)
- Soil—Categorized by surficial horizon
 - **Bold**—Peaty mud or peat
 - **Faint**—Dark mud
- Deposits between successive soils**
 - **Mud**
 - **Probable tsunami deposit on soil**
 - **Vented sand on soil**—Erupted from >35 m below land surface (Atwater, 1992, p. 1917)
 - **Muddy peat**—>0.3 m thick; records gradual submergence in the past 1000 years (Atwater and Moore, 1992)
- Vascular-plant fossils in growth position**
 - ▼ **Rhizomes in mud above soil**—*Triglochin maritimum* or, at South Fork, *Carex lyngbyei*
 - ▲ **Plant remains rooted in buried soil**—Herbs, shrubs, or trees
- **Microscopic fossils**—Diatoms at Redtail and Oyster localities (figs. 15, 16, and 20-23); diatoms and pollen at Johns River (Shennan and others, 1996)
- ~ **Unconformity on modern beach**
- Correlations**—Dashed where drawn through section in which soil is not recognized; queried where speculative. Letter gives name for soils inferred to correlate with one another
- Y — **Soil typically bold**
- W - - ? **Soil typically faint**

Ages of inferred earthquakes

Thick line gives 95-percent confidence interval calculated from error reported by laboratory; thin line increases this error by 1.6 or 2.0 (see text section "Radiocarbon Dating")

- **Closely limiting age**—Obtained from earthquake-killed plants (soils S, U, W, Y) or forest litter from uppermost 0.5 cm of buried soil (soils J, N); ages for soils S and Y denote inferred time of plant death
- ↓ **Minimum age**—Measured on rhizomes of *Triglochin maritimum* above soil
- ↑ **Maximum age**—Measured on woody root, plant detritus, or archaeological charcoal in buried soil
- ↑↓ **Minimum or maximum age**—Measured on peat in soil or on plant detritus in mud less than 20 cm above soil
- xxxx **Likely age range for inferred earthquake or earthquake series**

North America plate in western Washington 1,000–1,100 years ago. At that time, movement on one or more faults in the North America plate farther inland, in the Puget Sound area, generated 5–7 m of coseismic uplift near Seattle and several meters of coseismic uplift in an area 30–50 km to the southwest (Bucknam and others, 1992).

Despite this evidence for upper-plate seismicity, soil W at Willapa Bay probably represents an earthquake on the plate boundary. Radiocarbon-dated soils show that coseismic subsidence occurred elsewhere along the Washington coast between 900 and 1,300 years ago (Atwater, 1992, his figs. 6, 7)—at Neah Bay (location shown in fig. 1), at Grays Harbor along Elliott Slough, and at southern Willapa Bay along the Bear River (locations shown in fig. 5). We speculate that movement on faults and folds in the North America plate of western Washington accompanied—or shortly preceded or followed—one or more plate-boundary earthquakes 1,000–1,100 years ago, and that this plate-boundary seismicity is recorded at northeastern Willapa Bay by soil W.

SIZES OF THE EARTHQUAKES

Although their sizes are difficult to estimate, most or all of the earthquakes represented by buried soils H through Y were probably great earthquakes—of moment magnitude 8 or larger. Such magnitudes are suggested by geologic evidence for large ruptures and by a tradeoff between size and frequency.

DIMENSIONS OF PLATE-BOUNDARY RUPTURES

RUPTURE WIDTH

Wide ruptures are likely because, as shown in the section "Distinction Between Plate-Boundary and Upper-Plate Sources for the Earthquakes" (p. 80), the earthquakes marked by the buried soils probably occurred on the boundary between the Juan de Fuca plate and the North America plate. Ruptures on the Washington part of this plate boundary are likely to exceed 50 km in width, according to interpretation of geodetic and heat-flow data (Savage and others, 1991; Dragert and others, 1994; Hyndman and Wang, 1995). A potential rupture width of more than 100 km has been estimated from satellite measurements of deformation between 1992 and 1994 near the border between Washington and British Columbia (Dragert and Hyndman, 1995, p. 756).

High temperatures and fluid pressures may preclude ruptures much wider than 100 km on the Washington part of

Figure 32. Continued.

2. If the earthquake is followed by a long period without great earthquakes in the area, the buried soil that records the earthquake may be destroyed by decomposition after burial (p. 27; fig. 35).

Both these problems probably contribute to the spotty distribution and typical faintness of soil W. Little time may have elapsed between the earthquake marked by soil U (1,130–1,350 years ago) and soil W (900–1,300 years ago, perhaps 1,000–1,100 years ago). Soil W, where studied at the Oyster locality, had not yet become a high marsh at its time of subsidence and burial (p. 48). For many centuries thereafter the soil resided high in the profile of soil Y, where it was subject to decomposition before being lowered far into the intertidal zone about 300 years ago. This decomposition allowed destruction of soil W at the northwestern part of the Oyster locality (p. 45) and throughout the Redtail locality (p. 29).

We doubt, however, that any great earthquake of the past 3,500 years at northeastern Willapa Bay completely failed to leave a trace. Rather, we infer that a buried soil should be preserved, at least locally, for every earthquake that caused more than ½ m of widespread coseismic subsidence at northeastern Willapa Bay in the past 3,500 years—provided the soil had many decades to form.

MINIMUM TIME FOR FORMATION OF A PEATY SOIL

Nineteenth-century writings and maps suggest that 150 years is ample for reconstructing high marshes from muddy tidal flats at Willapa Bay. The writings come from one of the first Euro-American settlers of Willapa Bay, James G. Swan. Swan resided at the marshy mouth of the Querquelin (now Bone) River (fig. 6) in the early 1850's, barely 150 years after the most recent great earthquake in southern coastal Washington changed Willapa Bay's marshes into tidal flats. Though tinged with boosterism, Swan's account leaves little doubt that the bay's tidal marshes ("prairie-land") had already become high marsh in those 150 years:

"...all the rivers, large and small, run through fine prairie-land, exceedingly rich. That portion nearest the Bay is liable to be overflowed one or twice during the highest tides of winter, and are termed tide lands. This overflowing is, however, of no great detriment, although the water is salt, as, wherever the lands have been properly cultivated, they have yielded heavy crops. These prairies are all covered with grass of an excellent quality, making good grazing for stock, or a nutritious fodder when cut and made into hay." (Swan, 1857, p. 28).

Swan's account is consistent with depiction of tidal marshes on 1:10,000-scale plane-table maps of the bay's margins in the 1870's. These so-called T sheets, made by the U.S. Coast Survey, are on file with the National Oceanic and Atmospheric Administration in Rockville, Maryland; in addition they can be viewed and reproduced from microfilm at this agency's library in Seattle. The maps show tidal marshes similar in extent to those of modern Willapa Bay. Symbols on the maps further imply that most of these marshes stood about as high as they do today. The symbols include a solid line that separates marsh from mudflat except for a few small patches of marsh surrounded by mud-

flat. Whereas the patches of marsh consisted of *Triglochin maritimum* that was colonizing the mudflat in the 1870's, the solid line connotes the landward limit of ordinary high tides—tides that failed to cover most of the marshes.

EVENTS FOR WHICH SOILS WERE DESTROYED

Although centuries of decomposition may suffice to cause total destruction of shallowly buried soils at Willapa Bay, even 1,000 years is not enough for such destruction if the soil is more than about 0.3 m below the surface of the postearthquake marsh or swamp.

A postearthquake interval of 700–800 years is probably long enough to destroy a buried soil within 0.2–0.3 m of a high-marsh or spruce-swamp surface at northeastern Willapa Bay. Such an interval, as discussed on page 85, allowed destruction of soil W at part of the Oyster locality and throughout the Redtail locality.

Slightly deeper burial, however, allowed preservation of soil N during a postearthquake interval of about 1,000 years. The soil, which is barely visible where 0.2–0.3 m below soil S at the Jensen locality, is locally bold where 0.4 m below soil S at the Redtail locality (fig. 13, horizontal coordinate 18 m) and widely bold where 0.5 m below soil S at the Heron locality (fig. 25, horizontal coordinate 106–112). The interval during which soil N could have been destroyed by decomposition began soon after 2,400–2,800 years ago (the time of the event marked by soil N) and lasted until 1,500–1,700 years ago (the event marked by soil S).

Such minimum depths of burial to preserve a soil are probably smaller than the amounts of coseismic subsidence that the soil represents, for two reasons:

1. Amounts of subsidence along the Niawiakum River are probably 1 m or more for all but soil W, as inferred from the paleoecologic contrast between soil W and immediately overlying mud (p. 83).
2. Burial depth should be smaller than coseismic subsidence where interseismic uplift reduces the thickness of postseismic mud needed to rebuild a high marsh (p. 83). Interseismic uplift, which within 15 years undid as much as one-third of the 1964 subsidence in Alaska (Brown and others, 1977), should be the norm at northeastern Willapa Bay, particularly where soils cannot settle because they are underlain, at shallow depth, by scarcely compressible Pleistocene deposits. Pleistocene deposits at Willapa Bay record net late Quaternary uplift that cannot have been accomplished without interseismic uplift (p. 9).

INTERVALS BETWEEN GREAT EARTHQUAKES

The intervals between great earthquakes inferred from the buried soils can be portrayed simplistically as centuries of tectonic uplift and tidal sedimentation between instants of tectonic subsidence. We depict such history schemati-